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Performance of Public Transit Modes for Last Mile Connectivity near the Mass Rapid Transit System- a Case Study of Gurugram city

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Abstract

The last mile is defined as the movement of people from various transit stations to their destinations. The last mile transportation system provides on-demand shared transportation. The constraint in selecting the shared mode of transport arises when seeking better transportation services in terms of availability, flexibility, and affordability from various transit stations to the user destinations. The intermediate public transportation present at the various transit stations challenges private transit in providing a more flexible transportation mode to users with a reasonable cost parameter. The literature, based on the relative user characteristics of both private transportation and intermediate public transportation, requires a careful understanding of their mode choice travel behaviour. The paper explores the various socio-economic and travel characteristics that affect the last mile trips and examines the various issues encountered by users during the last-mile travel. The list of socio-economic and travel characteristics considered in the research conducted includes gender, age, occupation, household income, vehicle ownership, trip length, type of trip, total time, waiting time, and the reason for the mode choice. A questionnaire survey with responses from 1000 people was conducted at five metro stations in Gurugram city. The one-way ANOVA test and Chi-square test were applied to the collected data. The results show that the majority of variables among the socio-economic characteristics and travel characteristics are statistically significant at a 90% confidence interval, meaning that the socio-economic characteristics and travel characteristics of both private and intermediate public transportation users vary significantly across most variables. Pearson's correlation was conducted to check the multicollinearity of the variables. The binary logistic regression analysis shows that most trips taken were home-based trips. They can be classified as very short trips (<1km), short trips (1-5km), and medium trips (>5km – 10km). It was observed that variables such as gender, occupation, reason for mode choice, and total time have a significant influence on users when choosing their modes.

Keywords: Last Mile, Socio-Economic Characteristics, Travel Characteristics, Pearson's Correlation, ANOVA, Chi-Square test, Logistic regression analysis

1. Introduction

The last mile is the movement of individuals from a hub to a destination. Initially, it was used in the telecommunications and technology industries to describe the technologies and processes used to connect the end

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customer to a communications network. Last-mile connectivity enables commuters to quickly connect to the mainline, either by bus or perhaps a rail line. According to the Federal Transit Administration, the walking distance from the starting point, i.e., a hub to probably the nearest public transit system, should be within a one-mile radius. Biking access needs to be within a three-mile radius. As per the study, respondents use private vehicles and taxis from service providers in Beijing, leading to a reduction in the passenger vehicle ratio for comfort and reduced travel time (Hai Wang, 2014). Intermediate Para transit is generally preferred in areas near a predominant public transport nodal junction to provide last-mile connectivity. It is essential in Indian cities since they can be availed for short-haul trips by people for reaching their destinations at an affordable cost.

1.1 Public mode choice to connect the Public Transit system to the destination

Several studies have been conducted, considering the use of public transport modes and their relative distance from the last mile/first mile. The last mile problem deals with providing travel services from the workplace or home to the closest public transportation node (the first mile) or perhaps the other way around (the last mile). The public transportation service point should be nearest to the rapid transit rail station, mass rapid transit system, or a regular bus line. Unavailability of service typologies is a prime reason for discouragement among public transport commuters in urban spaces, especially school children, the elderly, differently-abled individuals, pregnant women, and so on (Joel Hansson, 2019). The latest survey conducted by the Ministry of Urban Development reflects that the number of people using feeder buses and two-wheelers as last-mile modes is comparatively lower than the number of people preferring Intermediate Public Transportation (IPT) for short distances. The study reveals that commuters prefer affordable modes of travel like auto-rickshaws, e-rickshaws, etc., for covering the last mile. Some services, such as ride & park and bike-sharing programs, serve as great provisions for the commuters (Smith, 2017). If supported by pedestrian and bicycle infrastructure provision, they may serve as practical options for public transport users with better accessibility (Active Transportation Alliance, 2014). The public transportation network should provide for the last mile to reduce the additional time and hassle of connecting from home to the road (Chong, 2011).

1.2 Objective of the study

To ascertain the user characteristics of both private transportation and Intermediate Public Transportation (IPT) during their ingress and egress trips and to identify the diverse factors influencing users' mode choice behaviour.

1.3 Literature Review

Intermediate public transportation (IPT) bridges the gap between private and formal public transport modes. IPT modes are mostly individually owned and operated, offering greater demand responsiveness than the formal bus system (Gadepalli, 2016). According to a primary survey conducted at various metro stations in Delhi city, 13% of commuters opt for auto-rickshaws per thousand population, especially in smaller cities, where the supply ranges from 0.3 to 2 auto-rickshaws per thousand population (BALAKRISHNAN, 2015). Intermediate public transit (IPT) plays a vital role in the global transportation system, catering to the mobility needs of billions of people daily. It primarily serves as a link between private and public transit in most locations. IPT modes, such as auto-rickshaws, are preferred by commuters due to their comfort and affordability (Aishwarya Jaiswal, 2022). IPT modes account for almost 50% of the total journey cost. While cycle rickshaws, considered the most eco-friendly feeder systems, are seldom used in specific parts of the city, they still serve 18% of total public transit commuters for last-mile connectivity (Chidambara, 2012).

In cities like Banaras, Kolkata, Siliguri, Vijayawada, battery-operated/electric rickshaws are now in use. According to the Ministry of Road Transport and Highways (MoRTH) report prepared by the Government of India ((MORTH), 2013), e-rickshaws should meet practical requirements, with a maximum seating capacity of four passengers, maximum luggage capacity of 40 kg, and a design speed of 15 km/hr. They should also be equipped with inbuilt GPS and CCTV tracking systems for safety, especially for female passengers. The design of e-rickshaws should prioritize safety and convenience, including the provision of fire extinguishers and first-aid kits. One can also consider facilitating an IPT system integrated with a public bicycle system (PBS) to provide low-carbon, environmentally friendly transport for short distances. This system is currently in practice in developed countries like Europe, North America, Asia, and Australia (Zhili Liu, 2012). Community-based bicycle sharing systems come in two types, one designed for community use and the other for residential use (Matsuura, 2003).

The socio-economic characteristics and travel patterns for both feeder buses and IPT modes differ. As travel time increases, people tend to prefer buses. Feeder buses have a lower floor height (650mm) with 27 seating and 20 standing capacities. However, only 25 such buses are operational (Annual Report, 2018). An analysis indicates that due to increased waiting times, only 2% of Delhi's population uses feeder buses. To encourage more public transportation users to opt for buses, the waiting period for feeder buses should be reduced (Ravi Gadepalli, 2020).

The primary goal of transit providers is to transport passengers safely, reliably, and conveniently between residential and commercial areas. Passenger safety is a top priority for transit systems (Pedestrian Safety Guide for Transit Agencies, 2008). Transit systems encompass public bus services, Bus Rapid Transit Systems (BRTS), railroads, vanpools, and more. These systems operate on streets, highways, or dedicated alignments and aim to retain existing riders while attracting new ones.

In the United States, residents of age-restricted neighborhoods who cannot operate personal vehicles may experience a lack of mobility. For them, options such as public transit and walking can be employed as modes of transportation (Michelle Oswald Beiler, 2016). Accessibility for pedestrians to transit systems is positively correlated with transit ridership (Hochmair, 2011). The Florida Department of Transportation introduced a variable element for assessing the transit level of service, which includes walking to transit stops (Wibowo, 2005). Emphasizing walking as the primary means of access for moving traffic is gaining serious consideration (Rongrong Yang, 2013).

Enhancing pedestrian access to public transportation and strengthening transit facilities to promote pedestrian traffic offer numerous benefits, including reducing the need for personal vehicles and mitigating environmental impacts such as greenhouse gas emissions and smog. Walking to and from public transportation helps maintain a healthy lifestyle, reduces traffic congestion effectively, and offers economically viable and environmentally friendly travel (Michelle Oswald Beiler, 2016). It also helps reduce total infrastructure costs. A survey conducted near Delhi metro stations shows that 47% of passengers use Intermediate public transportation, 17% use private vehicles, and 36% choose to walk (Chidambara, 2012). According to a 2015 survey by the Ministry of Urban Development, 59% of New Delhi residents opt for intermediate public transportation, while only 10% use private automobiles, and 27% prefer walking (ANNUAL REPORT, 2014-2015).

The research highlights the importance of last-mile transportation and the choice of transportation modes by users. It identifies the need to understand the mode choice behaviour of users concerning private transportation and intermediate public transportation (IPT). The research gap lies in the limited comprehensive studies in the literature that focus on relative user characteristics and their mode choice behaviour, particularly in the context of last-mile travel. This gap underscores the need for an in-depth examination of the socio-economic and travel characteristics affecting last-mile trips.

2. Methodology

Gurgaon, officially named Gurugram, is a satellite city of Delhi located in the north western part of the state of Haryana in India. It is the part of National Capital Region (NCR) of India. It is located 32 kilometres away from the southwest of New Delhi and 268 km (167 mi) south of Chandigarh, the capital city of the Haryana state. As of 2018, Gurgaon had a population of 2.1 Million. Gurgaon has turned into a top industrial and financial hub with the third highest per capita income in India. The city has a total area of 1258 square kilometres (District gurugram, 2018). However, the public transportation systems are poor and unmet considering the spike in the urban growth rate. The city has 142 buses which are lower than the urban bus standards set by the Ministry of Housing and Urban Affairs, and a metro line of 11.7 km intersecting with Yellow Line (consisting of 5 stations), insufficient to the demand. IPT is mainly disorganised and lightly governed by the Haryana government.

The average ridership at the five metro stations of Gurugram region was 3.27 lakhs per day as of 2021 (Detailed Project Report, 2019) . A pilot study of 20 samples each at five metro stations was conducted and the retention rate of the pilot study exceeds 90%. Hence the main study is feasible without changes to the protocol. All the questions considered in the questionnaire are viable for the main study. Based on Passenger ridership details provided by DMRC, a data from sample set of 200 people at five metro stations was used for the questionnaire survey. Hence a cumulative data set of 1000 samples were used for this research. The sample random sampling technique is used for the data collection. The 50% of data was collection from the female population at the each metro station. The proposed methodology requires detailed secondary data on the accurate ridership details to develop the exact sampling strategy. However, such detailed data is sometimes not available for cities in developing countries (RaviGadepalli, 2020). The questions covered during the survey are presented in the Annexure. The information about the commuters regarding their socio economic characteristics such as household income, number of members and vehicles ownership, information related to each traveller's age, gender, education, occupation and income along with their detailed travel behaviour (ANNUAL REPORT, 2014-2015) (Chidambara, 2012) (RaviGadepalli, 2020) . Several studies have indicated the use of socio economic characteristics in determining the travel behaviour of commuters opting IPT Modes. The list of socio-economic and travel characteristics considered in the questionnaire survey are shown in Table 1 and Table 2. Statistical tests were carried out to measure the variance between the IPT modes and private user groups for each of the variables. The chi Square test was used for categorical variables like gender, occupation, income range and vehicle ownership, while the ANOVA was used for age, which is a continuous variable. The logistic-regression analysis with mode choice as the dependent variable and user characteristics as independent variables was carried out to identify the key variables impacting user choice between IPT and private modes. It will help the policy makers and planners in understanding the user behaviour in a detailed manner by enhancing the transit ridership (RaviGadepalli, 2020). The approximate ridership details shown below at each metro station in the city were provided by the DMRC (Delhi Metro Rail Corporation Limited) (Detailed Project Report, 2019).

- HUDA City Centre (Approx.80,000 per day)
- IFFCO Chowk (Approx.40,000 Passengers per day)
- MG Road (Approx.60000 Passengers per day)
- Sikanderpur (5 lakh Passengers per day)
- Guru Droncharya. (Approx. 20,000 Passengers per day)

3 Data Collection

At Huda City Centre metro station, it was observed that 94% of commuters used Intermediate public transportation (IPT), while 6% traveled by private transportation. The available IPT modes at Huda City Centre metro station included e-rickshaws (informal transportation), contract carriages, shared autos (informal transportation), feeder buses (informal transportation), and contract carriage cabs (informal transportation).

At IFFCO Chowk metro station, 90% of travelers utilized Intermediate public transportation, 5.5% opted for private vehicles, and 4.5% commuted by walking. The IPT modes available at IFFCO Chowk metro station included cycle rickshaws (informal transportation), contract carriages, shared autos (informal transportation), feeder buses (informal transportation), and contract carriage cabs (informal transportation).

At MG Road metro station, 77% of commuters used IPT modes, and 23% walked to reach their destination. The IPT modes available at MG Road metro station comprised e-rickshaws (informal transportation), contract carriages, shared autos (informal transportation), feeder buses (informal transportation), and contract carriage cabs (informal transportation).

At Sikanderpur metro station, 93.5% of commuters relied on IPT modes, 4% used private transportation, and 2.5% walked to their destinations. The IPT modes used at Sikanderpur metro station included e-rickshaws (informal transportation), contract carriages, shared autos (informal transportation), feeder buses (informal transportation), and contract carriage cabs (informal transportation).

At Guru Dronacharya metro station, 92% of commuters used IPT modes, 6% chose private transportation, and 2% walked to their destinations. The IPT modes available at Guru Dronacharya metro station consisted of contract carriages, shared autos (informal transportation), feeder buses (informal transportation), and contract carriage cabs (informal transportation).

The list of socio-economic characteristics was provided in Table 1, and the list of travel characteristics was displayed in Table 2.

Table 1 List of socio-economic characteristics

S NO	Socio-Economic Characteristics	
	Short-form	Full form
1	G	Gender
2	A	Age
3	O	Occupation
4	HI	Household Income
5	VO	Vehicle Ownership
6	s	student
7	w	working
8	r	retired
9	h	housewife
10	M	Male Population
11	F	Female Population

Table 2 List of Travel characteristics

S NO	**Travel Characteristics**	
	Short-form	**Full form**
1	TTL	Total trip length
2	TOT	Type of trip
3	WT	Waiting Time

4	TT	Total Time
5	RMC	Reason for mode choice
6	AVL	Available
7	FL	Flexible
8	RF	Reasonable fare
9	A	All

Table 3 Percentage of mode share based on variable classification

	GROUPS	PRIVATE	IPT	TOTAL
sample size		43	894	937
Gender	Male	78%	85%	85%
	Female	22%	15%	15%
Age	<18	0%	2%	2%
	18-40	91%	84%	84%
	41- 60	9%	11%	11%
	>61	0%	3%	3%
Occupation	student	9%	15%	15%
	working	79%	80%	81%
	retired	0%	3%	3%
	housewife	12%	2%	1%
Household Income	0-2.5	2%	1%	1.00%
	2.5-5	63%	67%	68%
	>5 - 10	31%	25%	26%
	>10	4%	7%	5%
Vehicle ownership	2w	51%	65%	64%
	car	7%	2%	2%
	other	3%	1%	0.3%
	2w+cars	16%	17%	18%
	2w+c+o	7%	2%	2%
	2w+others	16%	12%	13%
	Cars+others	0%	1%	0.21%
Trip length	<1	2%	1%	5%
	1-5	40%	28%	29%
	>5 – 10	41%	36%	35%
	> 10	17%	35%	32%
Type of Trip	Home Based	81%	78%	79%
	Non-Home based	19%	22%	21%
Time	<15	67%	43%	47%
	15-30	30%	38%	36%
	30-60	0%	14%	13%
	>60	3%	5%	4%
Wait Time	0-10	74%	79%	80%
	>10-15	0%	1%	1%
	>15	0%	0%	0%
	No, wait	23%	20%	21%
Reason for mode	Available	86%	41%	41%
	Flexible	2%	5%	5%
	Reasonable fare	0%	7%	7%
	All	12%	47%	48%

From Table 3, it is evident that the male population exhibited a greater preference for IPT over private vehicles compared to the female population. Commuters between the ages of 18 to 40 were found to make a higher number of

last-mile trips. Working commuters made more trips than students, retired individuals, and homemakers. It was noted that the household income of last-mile transport users predominantly fell within the range of earning 2.5 to five lakhs per annum, and they primarily used IPT modes for their last-mile travel. Users with an annual income between 5 lakhs to 10 lakhs per annum showed a stronger inclination towards private transportation. Additionally, it was observed from the collected samples that each family maintains at least one two-wheeler.
Private vehicle users tended to undertake the majority of their trips with a trip length ranging from 1 to 10 kilometres, while IPT modes had a predominant share for trip lengths greater than 10 kilometres. Most trips taken by travellers in the region were home-based trips. Commuters in the region equally favoured both private and IPT modes for their last-mile transportation. Specifically, 67% of private vehicle users were observed to complete trips in less than 15 minutes to reach their respective destinations, whereas 38% of IPT mode users were observed to require 15 to 30 minutes. Furthermore, it was noted that 80% of commuters were unwilling to wait for the feeder bus for more than 10 minutes. The primary reason travellers chose Intermediate public transportation was due to its consistent availability and flexibility, combined with reasonable fares.

4 Data Analysis

The data analysis for collected questionnaire data was carried out by making use of Statistical Package for the Social Sciences (SPSS) Software.

4.1 Results and Discussions of data Analysis

The one-way ANOVA test and Chi-square test in Table 4 and Table 5 show that the majority of variables related to socio-economic and travel characteristics are statistically significant at a 90% confidence interval. In other words, the socio-economic characteristics and travel patterns of both private and IPT mode users significantly differ across most variables. Individuals from diverse socio-economic backgrounds and with varying travel characteristics utilize the available private and IPT modes.

Table 4 Chi-square and ANOVA (P – Values) of socio-economic characteristics

Variable(s)	GROUPS	CHI SQUARE TEST	ANOVA TEST
Gender	Male Female	0.518	
Age	<18 18-40 41- 60 >61		0.357
Occupation	student working retired housewife	0.076	
Household Income	0-2.5 2.5-5 >5 - 10 >10		0.030
vehicle ownership	2w car Others	0.006	

Table 5 Chi-square and ANOVA (P – Values) of travel characteristics

Variable(s)	GROUPS	CHI SQUARE TEST	ANOVA TEST
Trip length (KM)	<1 1-50 >5 - 10 > 10		0.037
Type of Trip	Home Based Non-Home based	0.630	
Time (Minutes)	<15 15-30 30-60 >60		0.05
Wait Time (Minutes)	0-10 >10-15 >15 No wait		0.815
Reason for mode	Available Flexible Reasonable fare All	0.000	

A correlation analysis was conducted to examine multicollinearity among the socio-economic and travel characteristics, as depicted in Table 6 and Table 7, respectively. In Table 7, among all the travel characteristics, the correlation between total time and total trip length is notably positive and significant. In other words, when the overall trip length increases, the total time required to complete the trip also increases.

Table 6 Correlation matrix for socio-economic characteristics

		G	VO	HI
G	Pearson Correlation	1	.040	-.077*
	Sig. (2-tailed)		.219	.018
	N	1000	1000	1000
VO	Pearson Correlation	.040	1	.208**
	Sig. (2-tailed)	.219		.000

	N	1000	1000	1000
HI	Pearson Correlation	-.077*	.208**	1
	Sig. (2-tailed)	.018	.000	
	N	1000	1000	1000

Table 7 Correlation matrix for Travel characteristics

		TOT	WT	TTL	RMC	TT
TOT	Pearson Correlation	1	.001	.305**	.088**	.341**
	Sig. (2-tailed)		.985	.000	.007	.000
	N	1000	1000	1000	1000	1000
WT	Pearson Correlation	.001	1	.001	.023	.007
	Sig. (2-tailed)	.985		.970	.482	.822
	N	1000	1000	1000	1000	1000
TTL	Pearson Correlation	.305**	.001	1	.165**	.716**
	Sig. (2-tailed)	.000	.970		.000	.000
	N	1000	1000	1000	1000	1000
RMC	Pearson Correlation	.088**	.023	.165**	1	.216**
	Sig. (2-tailed)	.007	.482	.000		.000
	N	1000	1000	1000	1000	1000
TT	Pearson Correlation	.341**	.007	.716**	.216**	1
	Sig. (2-tailed)	.000	.822	.000	.000	
	N	1000	1000	1000	1000	1000

**. Correlation is significant at the 0.01 level (2-tailed).

The relative impact of socio-economic and travel characteristics on users' mode choice between private and IPT modes was determined through logistic regression analysis, as presented in Table 8. In this analysis, the mode choice served as the dependent variable, while various socio-economic and travel characteristics were considered as independent variables.

For the logistic regression, private vehicles were coded as 1, and Intermediate public transportation (IPT) modes were coded as 2. Male gender was represented as 1, while female gender was coded as 2. Variables like occupation, type of reason, and trip purpose were treated as categorical variables. Age, household income, total trip length, waiting time, and total time were considered continuous variables. The logistic regression model exhibited an R-squared value of 0.617, indicating the model's suitability.

Among the socio-economic variables, the occupation types "working" and "retired people" displayed a significant positive correlation. This suggests that working individuals prefer both private and IPT modes, while retired individuals lean more towards IPT modes than private transportation.

Among all the travel characteristics, the type of reason, such as "availability" and "flexibility" of transportation, exhibited a significant correlation. The negative correlation suggests that many working individuals choose private vehicles because they have the option of either cars or two-wheelers at their homes, and they perceive private vehicles as more flexible than intermediate public transportation.

In terms of travel characteristics, total time demonstrated a significant positive correlation, signifying that the total time taken by a car depends on the total trip length. For longer trips (>10 km), people tended to use intermediate public transportation, including feeder buses and non-shared cabs, as their mode choice. For medium trips (>5 - 10 km), intermediate public transportation options such as contract carriage autos and non-shared cabs were preferred. For shorter trips (>1 - 5 km), people typically used intermediate public transportation, including e-rickshaws and shared autos.

Table 8 Logistic regression analysis for private vehicles and Intermediate public transport in Gurugram city

		B	Sig.
G	M	.071	.884
	F	0.34	0.04
O	s		.272
	w	2.123	.055
	r	1.919	.058
	h	19.29	.998
HI		.329	.353
A		.142	.789
TOT		.428	.326
RM C	AV L		.000
	FL	2.302	.000
	RF	1.3	.134
	A	16.46	.997
TT		.675	.063
WT		-.079	.553
TTL		.188	.511
C		14.61	1.00

5 Discussion and Conclusion

This paper offers an overview to understand Intermediate public transportation and private transportation in the developing city of Gurugram. It outlines the complete methodology for data collection and analysis of user characteristics for both personal and IPT modes. A questionnaire survey was conducted at five metro stations in Gurugram to collect these user characteristics, which are presented in the paper.

It was observed that the female population tends to take fewer trips in IPT when compared to the male population. Generally, the female population prefers safer and more convenient modes of travel, especially during the night. In Gurugram, the feeder buses and shared autos were often overcrowded and lacked public safety measures. Additionally, there was no ticketing facility inside the feeder bus. In some instances, contract carriage autos were used as shared autos when there was a higher influx of passengers near metro stations. More student trips were made using Intermediate public transportation, while working trips were taken using private and intermediate public transportation due to time constraints.

Based on people's opinions, it was noted that private two-wheeler transportation would take less time than IPT in reaching the metro stations during peak traffic hours. Metro stations like Huda City Centre and IFFCO Chowk provide ample space for parking private vehicles. The survey revealed that nearly every family owns a two-wheeler, indicating the prevalence of private transportation in the city.

Pearson's Correlation analysis showed that total trip time (TT) was dependent on waiting time and total trip length (TTL), while total trip length (TTL) was influenced by the type of trip (TOT). Most trips taken in Gurugram were home-based trips, meaning that the majority of last-mile trips were very short (< 1 km), short (1-5 km), and medium (5 km - 10 km) in length.

Policies should be implemented to make IPT modes more accessible and user-friendly. Dedicated feeder services can enhance the usability of these modes. For very short trips, walking is a preferred choice, while e-rickshaws and shared autos are popular for shorter trips. For medium trips, contract carriage autos and cabs are preferred. Intermediate public transport is attractive for passengers traveling short distances, and quality attributes such as frequency, comfort, reliability, travel time, and network coverage are important to regional travellers.

For longer trips (> 10 km), many people in the study area prefer feeder buses, but some are turning to cabs and private transportation due to safety concerns and overcrowded buses. To enhance the attributes of accessibility and serviceability, it's essential to make the feeder bus system a more formal part of public transport.

Public bike-sharing systems were launched at the Huda City metro station with good intentions but failed to attract significant ridership due to various reasons, including the need to return cycles to the same station, poor maintenance, and lack of technology for tracking bicycles and users. Implementing policies to ensure accountability for the public bike-sharing system may improve its usability.

To further this research, a network plan for an integrated bus system needs to be developed for Gurugram, especially for longer trips. Comparing Gurugram's data with user characteristics from other cities with similar modes of transportation can offer valuable insights. Analyzing the various characteristics of these modes and their usage in the city can provide valuable insights for policy, planning, and regulatory measures to enhance formal and informal public transport systems in Indian cities.

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Appendix A

Survey Questionnaire

Gender: ☐ Male ☐ Female

Age Group: ☐ <18 ☐ 18-40 ☐ 41-60 ☐ >60

Occupation: ☐ Student ☐ Working ☐ Retired ☐ Housewife

Household Income: ☐ 0-2.5L ☐ 2.5-5L ☐ 5-10L ☐ >10L

Vehicle Ownership:

Cars: ☐ 0 ☐ 1 ☐ 2 ☐ 3+ Others:____________________________

2W: ☐ 0 ☐ 1 ☐ 2 ☐ 3+

What is your last-mile and TL (Km):______________________________________?

How much time do you spend in commuting to your destination from metro/ railway station/ bus stand?

☐ <15 minutes ☐ 15-30minutes ☐ 30-1hour ☐ >1hour

How do you get to transit station? Cost of travelling?

Private vehicles: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

Non-shared cabs: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

IPT: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

Walk: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

(Name of the mode:______________) Actual Fare:_____________________

How do you get from the transit station to our destination? Cost of travelling?

Private vehicles: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

Non-shared cabs: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

IPT: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

Walk: ☐ Available ☐ Flexible ☐ Reasonable fare ☐ All

(Name of the mode:______________) Actual Fare:_____________________

Are you willing to use Public Bicycle sharing system? ☐ Yes ☐ No

Why? ____

How long are you willing to wait for public feeder buses?

☐ 0-10 ☐ 10-15 ☐ >15 ☐ No waiting

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